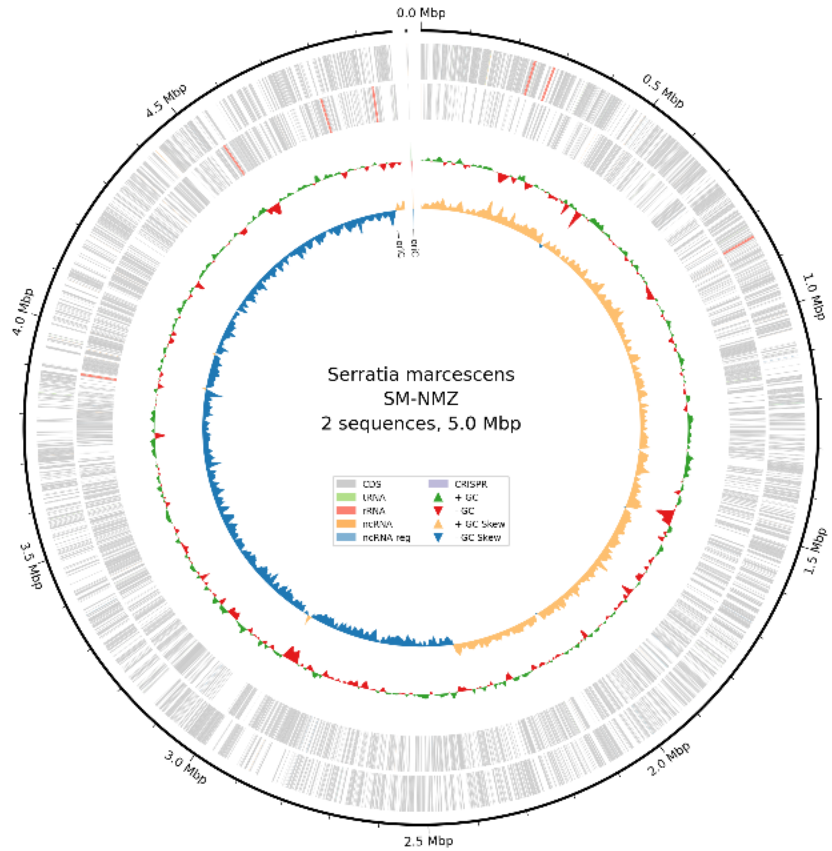
NMZ Serratia genomics

An explainer for the research group

SM-NMZ Slaughterhouse effluent
C14-NMZ Calotropis rhizosphere



21 October 2026

DNA, gene predictions and activity

Stage	What it supplies
Sequencing	DNA fragments called reads
Assembly	Reconstructed chromosome and other replicons
Annotation	Predicted genes and possible functions
Experiment	An observed effect under stated conditions

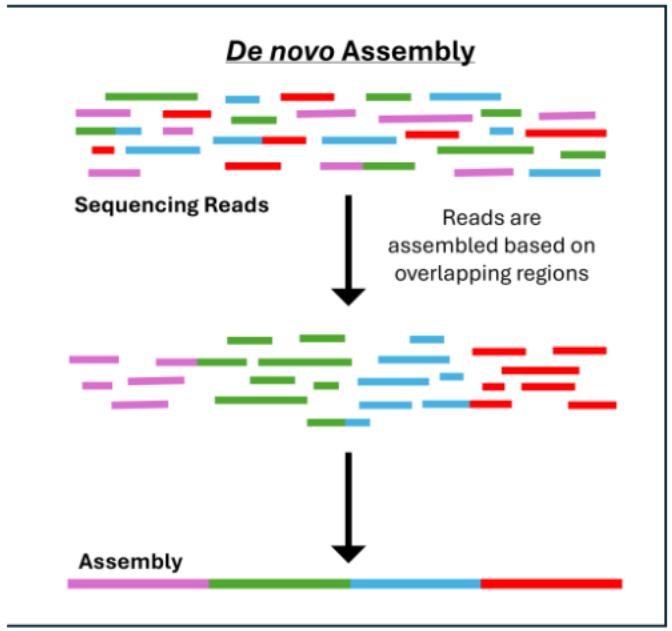
A predicted function is a candidate for explanation.

Reads, assembly and genome annotation

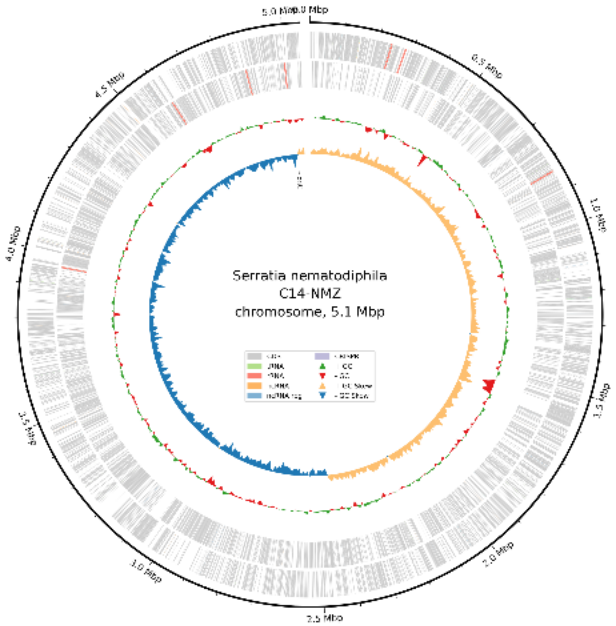
Basecalled reads → Draft assembly → Annotated genome

```
@SRR014849.1 EIXKN4201CFU84 length=93
GGGGGGGGGGGGGGGGCTTTTTTTGTTTGGAACCGAAAGG
GTTTTGAATTTCAAACCCTTTTCGGTTTCCAACCTTCCAA
AGCAATGCCAATA
+SRR014849.1 EIXKN4201CFU84 length=93
3+&$#""""""""""""7F@71,'";C?,B;?6B;:EA1EA
1EA5'9B:?:#9EA0D@2EA5':>5?:%A;A8A;?9B;D@
/=<?7=9<2A8==
```

DNA bases + quality



Overlaps → contigs



Features + predictions

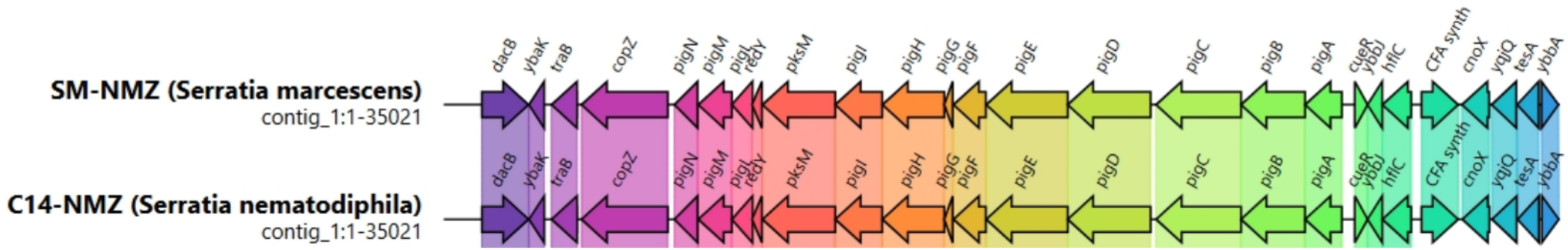
External teaching examples (left/centre); project Bakta output (right)

What each kind of evidence answers

Evidence	Question it addresses
Related gene or protein	Is there a plausible candidate?
Several linked genes	Does the proposed machinery fit together?
Chemical analysis	Which molecular assignment has support?
Activity assay	What effect occurred in this experiment?
Controlled perturbation	Does this factor contribute to that effect?

Genes, neighborhoods, clusters and operons

Term	Meaning
Gene	DNA sequence encoding a product
Neighborhood	Nearby genes: physical context
Biosynthetic cluster	Nearby genes implicated in making a molecule
Operon	Genes transcribed together as a unit



pig: a multi-gene example. SwrW: a multifunctional enzyme encoded by one gene.

SwrW evidence in the NMZ study

Observation or test	What it would establish
SwrW sequence, domains and context	A plausible biosynthetic candidate
Reported LC-HRMS assignments	Chemical support for serratamolides
Tension or E24 assay	Effect of the particular material tested
Controlled intervention / purified compound	A test of contribution to that effect

NMZ genomic and reported chemical evidence are available.

An activity result and a causal mechanism need their own evidence.

Genome resources and identity

Feature	SM-NMZ	C14-NMZ
Chromosome	4.971 Mb	5.051 Mb
Additional replicon	3.581-kb plasmid	None recovered
Bakta CDSs	4,566	4,626
GTDB r232 label	S. marcescens	S. nematodiphila

TYGS: one species cluster, separate subspecies clusters

ANI and digital DNA–DNA hybridization

Measure	What is compared	How it is interpreted
ANI + aligned fraction	DNA sequence identity in aligned regions	Identity together with alignment coverage
dDDH	Sequence-based estimate of DNA–DNA relatedness	Estimate and confidence interval

ANI = average nucleotide identity; AF = aligned fraction

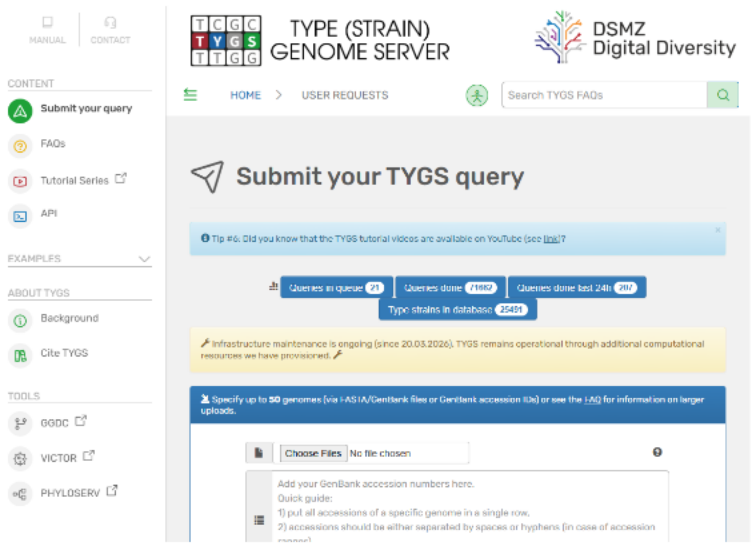
Broad species guides: ANI ≈95–96%; dDDH ≈70%

GTDB applies a representative-specific ANI radius.

NMZ pair: dDDH(d4) 74.2% (71.2–77.0%)

GTDB-Tk and TYGS workflows

TYGS · web service



Genome FASTA → type-strain comparisons
GBDP → dDDH + genome phylogram

GTDB-Tk · local pipeline

GTDB = Genome Taxonomy Database

Genome FASTA + GTDB references
↓
ANI + aligned fraction
↓
Representative-specific species boundary
↓
Classification summary

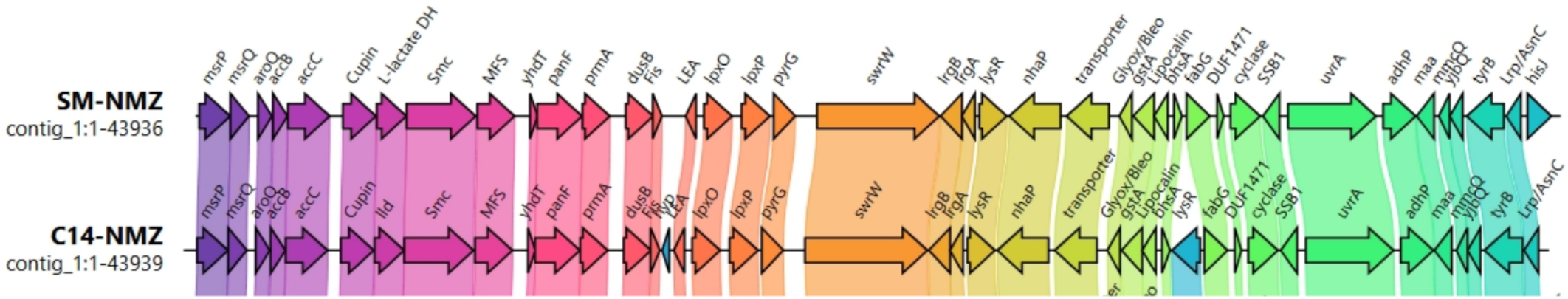
NMZ queries were classified by ANI screening.

C14: GTDB S. nematodiphila; TYGS S. marcescens

SwrW and reported serratamolides

Genomic evidence

Both isolates carry a conserved swrW candidate.



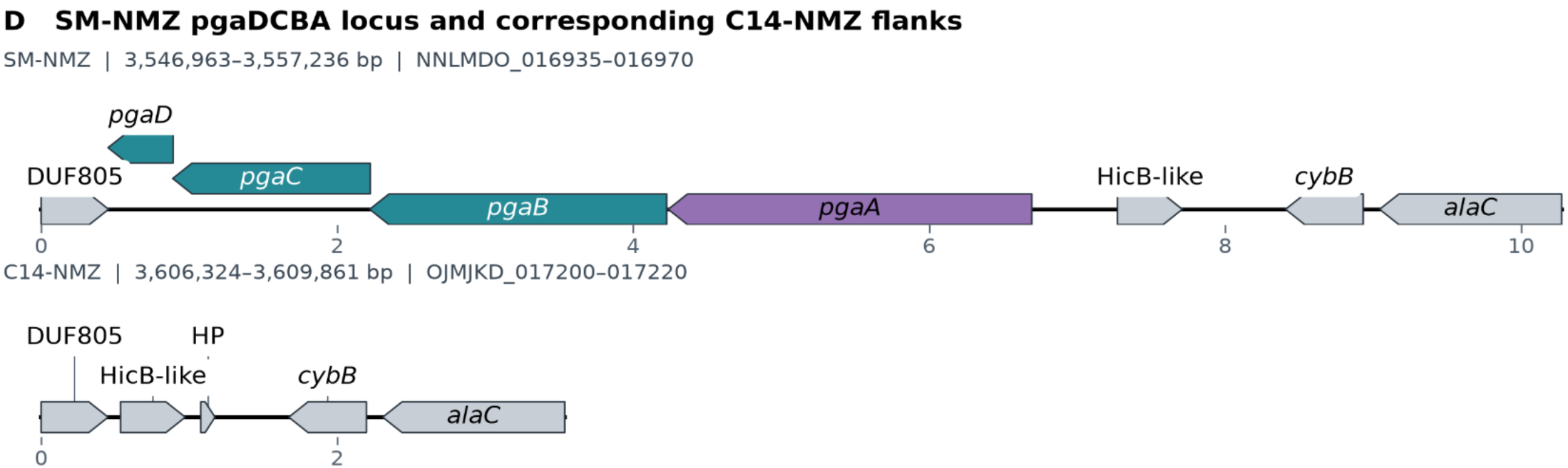
Reported chemical evidence

LC-HRMS: serrawettin W1 and serratamolide C

The pig locus also matches the red/pink culture phenotype.

Extracellular matrices and surface activity

Both isolates: capsule/EPS and cellulose candidates



Gene arrows. Scale in kb from the first displayed CDS.

SM has the contiguous pga locus associated with PNAG.

Matrix contribution depends on the material actually assayed.

EPS, CPS and matrix polysaccharides

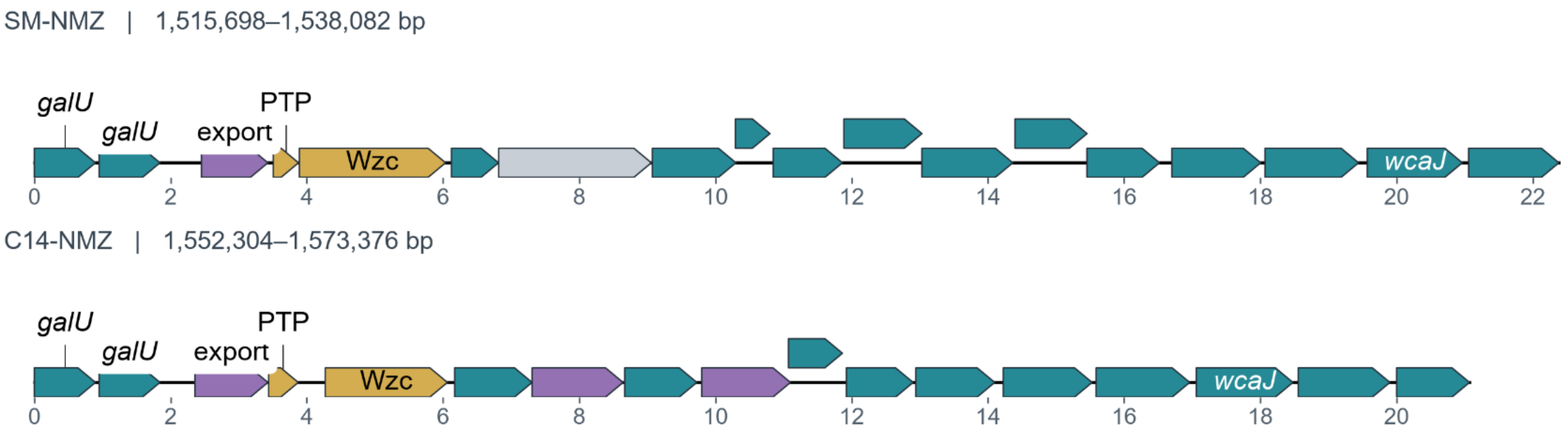
Polysaccharides are long chains of sugar units.

Term	What it means
EPS	Extracellular polymeric substances: the matrix
CPS	Capsular polysaccharide: associated with the cell
PNAG	β-(1→6)-linked N-acetylglucosamine polymer
Cellulose	β-(1→4)-linked glucose polymer

PNAG and cellulose have different sugar units and linkages.

A marker category does not identify the polymer in an NMZ sample.

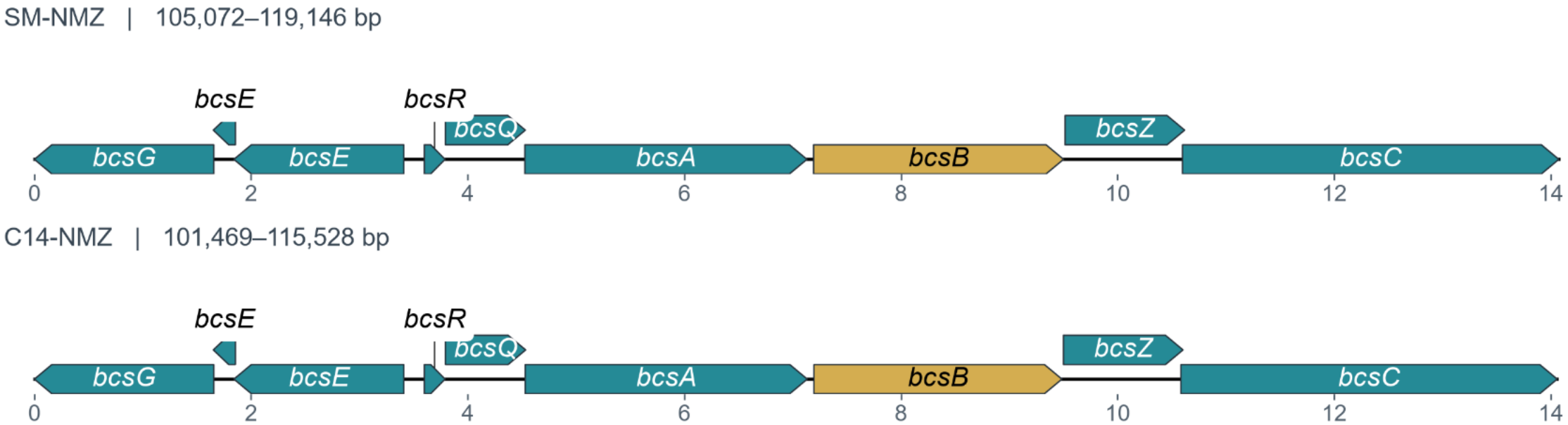
Capsule loci and the HADEG markers



PTP: tyrosine phosphatase · Wzc: tyrosine kinase

The HADEG “Emulsan” label does not identify the polymer product.

Cellulose-associated loci in both isolates

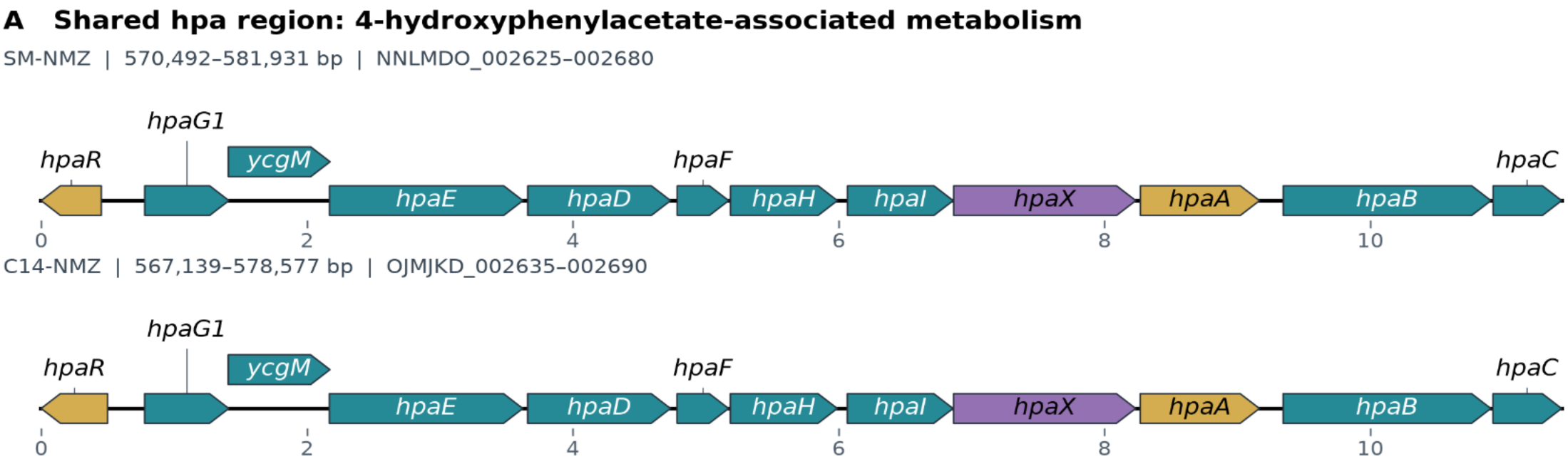


BcsA: synthesis · BcsB: partner · BcsC: export-associated

Scale: kb. Conserved gene order supports shared biosynthetic potential.

Aromatic metabolism depends on the substrate

Both isolates: organized hpa and paa loci



Gene arrows. Scale in kb from the first displayed CDS.

SM adds hydroxybenzoate/pca and gentisate candidates.

These findings concern specific oxygenated aromatics.

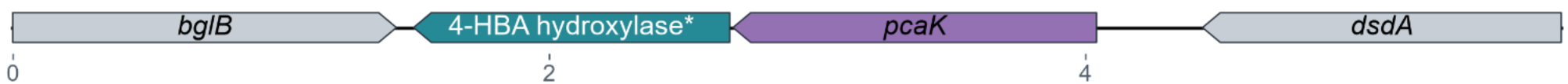
The HADEG Toluene_degradation hit

A characterized reference route · P. mendocina KR1

toluene → p-cresol → 4-HBA → protocatechuate

Initial activation and processing precede the PobA step.

SM-NMZ | 2,461,766–2,467,540 bp



SM NNLMDO_011765: a downstream hydroxylase candidate

4-HBA = 4-hydroxybenzoate. *Label abbreviates the source product.

LadA and the alkane-oxidation hypothesis

Historically proposed role

Long-chain alkane → primary alcohol

LadA was reported as a soluble, flavin-dependent candidate for this first step.

Why search here?

To identify a plausible initiating enzyme alongside downstream carbon metabolism.

LadA evidence across the literature

Study	Evidence reported	Interpretation
Feng et al. 2007	Canonical LadA: in-vivo and in-vitro support	Initial long-chain oxidation proposed
Boonmak et al. 2014	B23 homologs: alkane depletion in a heterologous host	Cell-based evidence; not purified enzyme
Mireles et al. 2025	No direct hydroxylation detected for tested proteins	First-step assignment questioned

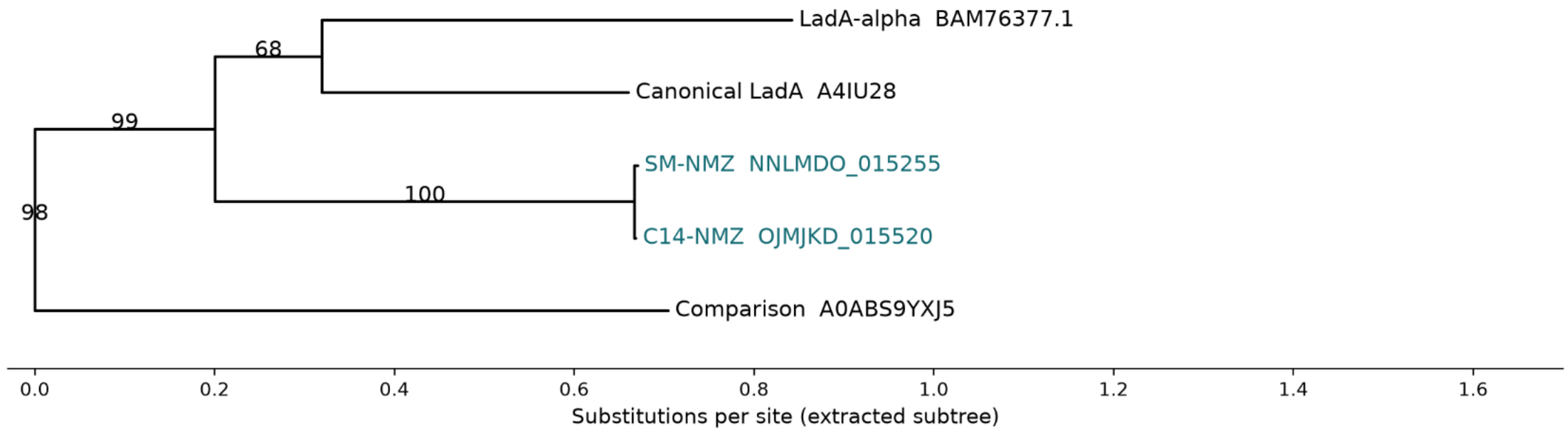
The NMZ homologs still need a direct functional assignment.

Initial alkane oxidation

Simplified step	Genomic interpretation
Alkane to alcohol	Initiating system unresolved
Alcohol to aldehyde	Downstream processing candidates
Aldehyde to fatty acid	Downstream processing candidates
Fatty acid to cellular metabolism	Beta-oxidation and assimilation context

DNA-repair AlkB and alkane hydroxylase AlkB are different proteins.

The LadA-like comparison



Node labels show UFBoot support (%).

Strong sequence grouping supports a candidate relationship.

Its substrate remains unresolved. Nearby genes add sulfur/methionine context.

Matching assays to genomic candidates

Endpoint, if measured	Relevant interpretation	Evidence to pair with it
Surface/interfacial tension	Amphiphile contribution	Material, interface, mN/m, control
E24 emulsification index	Amphiphile and/or matrix contribution	Oil, 24-h protocol, layer heights
Aromatic conversion	Route matching the exact substrate	Substrate, products and recovery controls
Pour point	Low-temperature oil-flow performance	Oil, dose, method and temperature

Numerical endpoint data remain with the experimental team.

Host-interaction genes

Reference-associated system	Biological role in characterized systems
ShlA–ShlB	Hemolysin machinery
Has	Heme acquisition
Type VI secretion system	Bacterial competition

Reference matches do not quantify pathogenicity or establish safety.

The genomics contribution to the paper

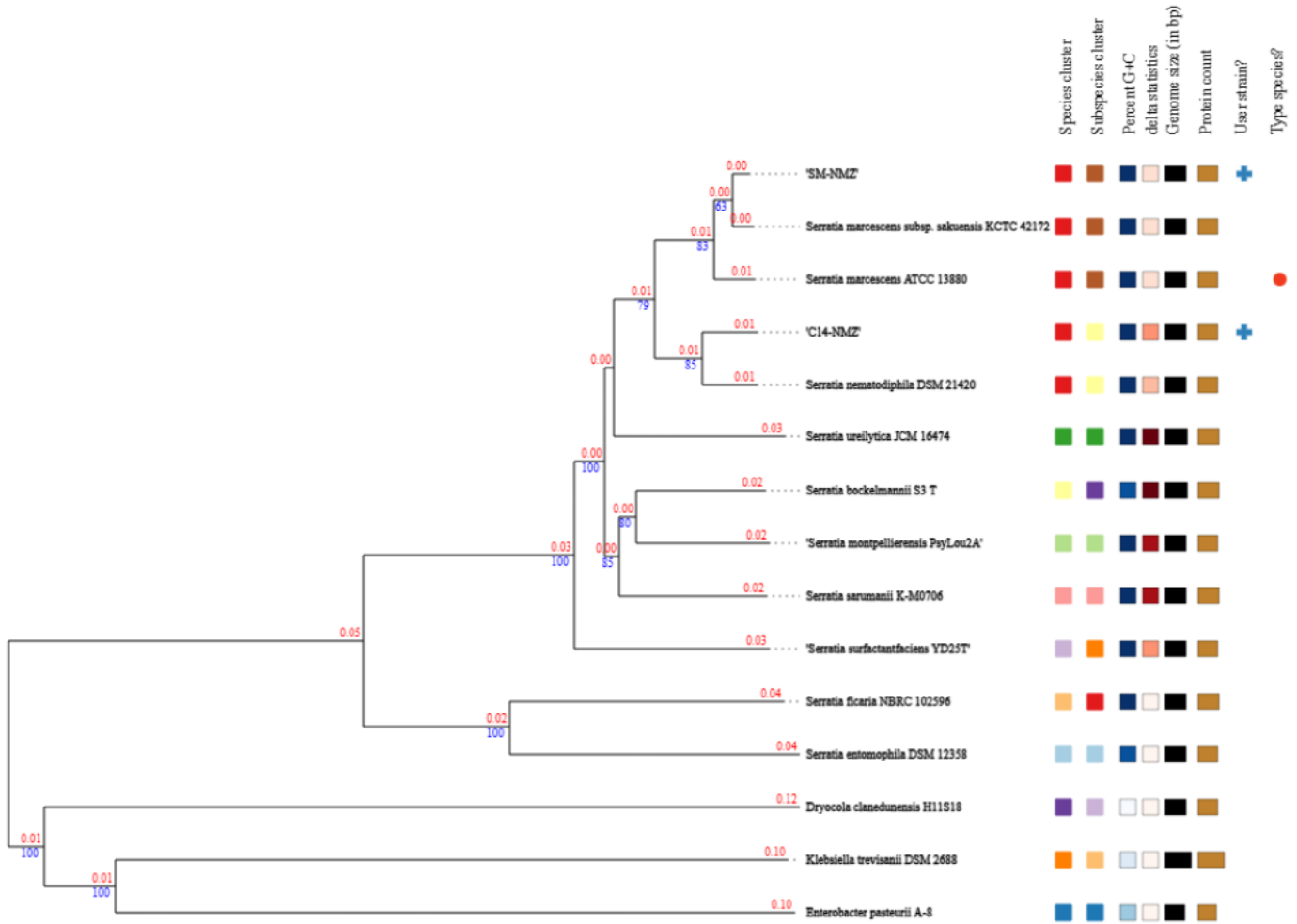
Finding	Contribution
SwrW and reported serratamolides	Specific surface-active metabolite lead
pig locus and red/pink cultures	Pigment-associated finding
Matrix loci and SM pga cluster	Shared systems plus a focal difference
Aromatic loci and LadA-like comparison	Substrate-specific context and an unresolved enzyme

Which measured endpoint and material should lead the paper?

Workflow and tools by purpose

Question	Tools and evidence
Which read evidence enters assembly?	NanoPlot, Chopper, Filtlong
How are sequences reconstructed?	Autocycler/assemblers, dnaapler, Dorado
Where are genes and what are candidates?	Bakta, antiSMASH, DRAM, CANT-HYD, HADEG
How are genomes and proteins related?	TYGS, GTDB-Tk, ProteinOrtho/PoFF, IQ-TREE
How are local arrangements compared?	clinker and DNA Features Viewer

The TYGS genome phylogram



Blue labels: pseudo-bootstrap support (%) from 100 replicates.

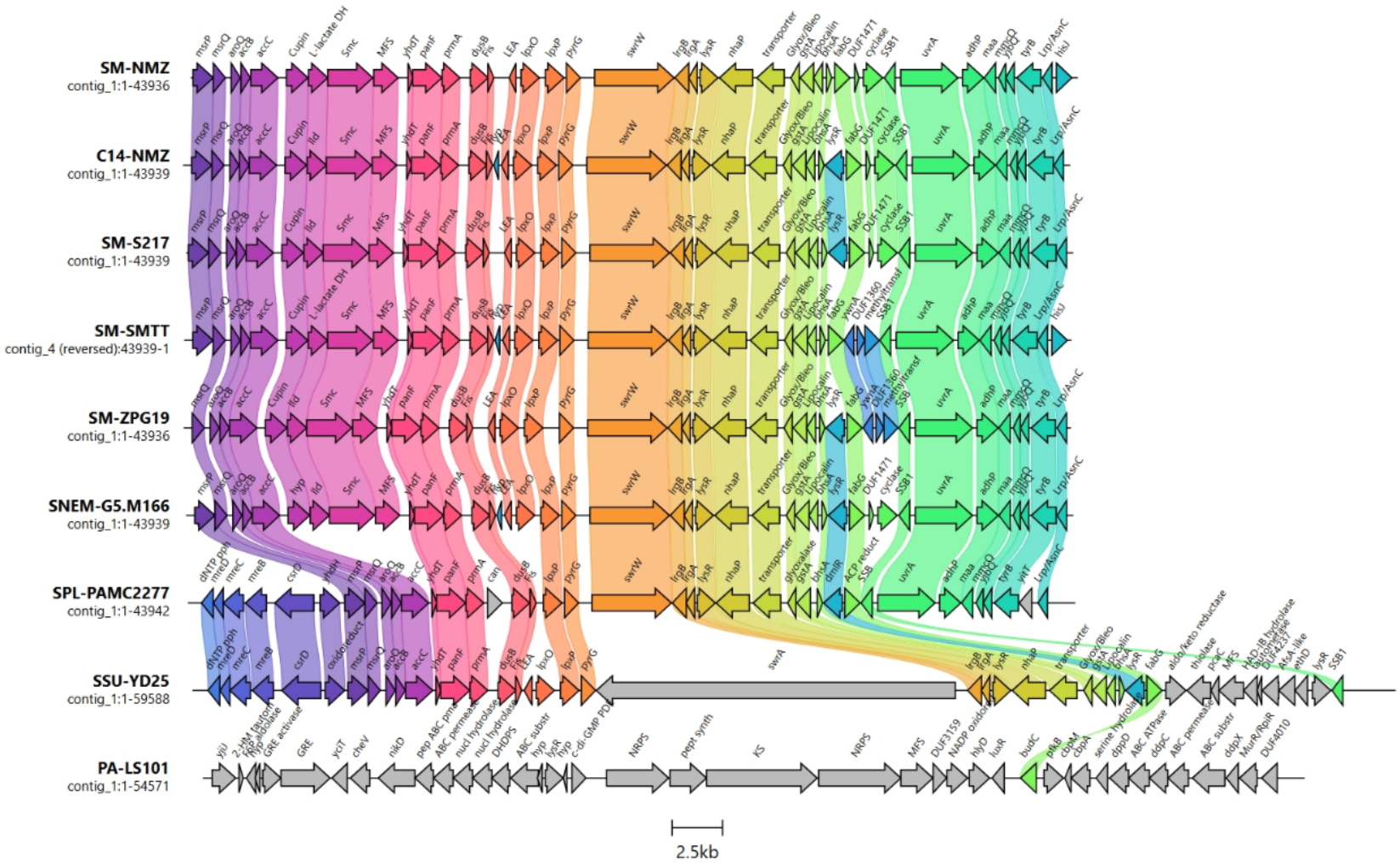
Taxonomic comparisons

dDDH(d4), %	SM-NMZ	C14-NMZ
S. marcescens ATCC 13880	90.7	73.7
S. nematodiphila DSM 21420	74.7	85.5
Direct focal comparison	74.2 (71.2–77.0)	Same comparison

GTDB ANI screening

SM 98.86% ANI (AF 0.947)
C14 98.22% ANI (AF 0.932)

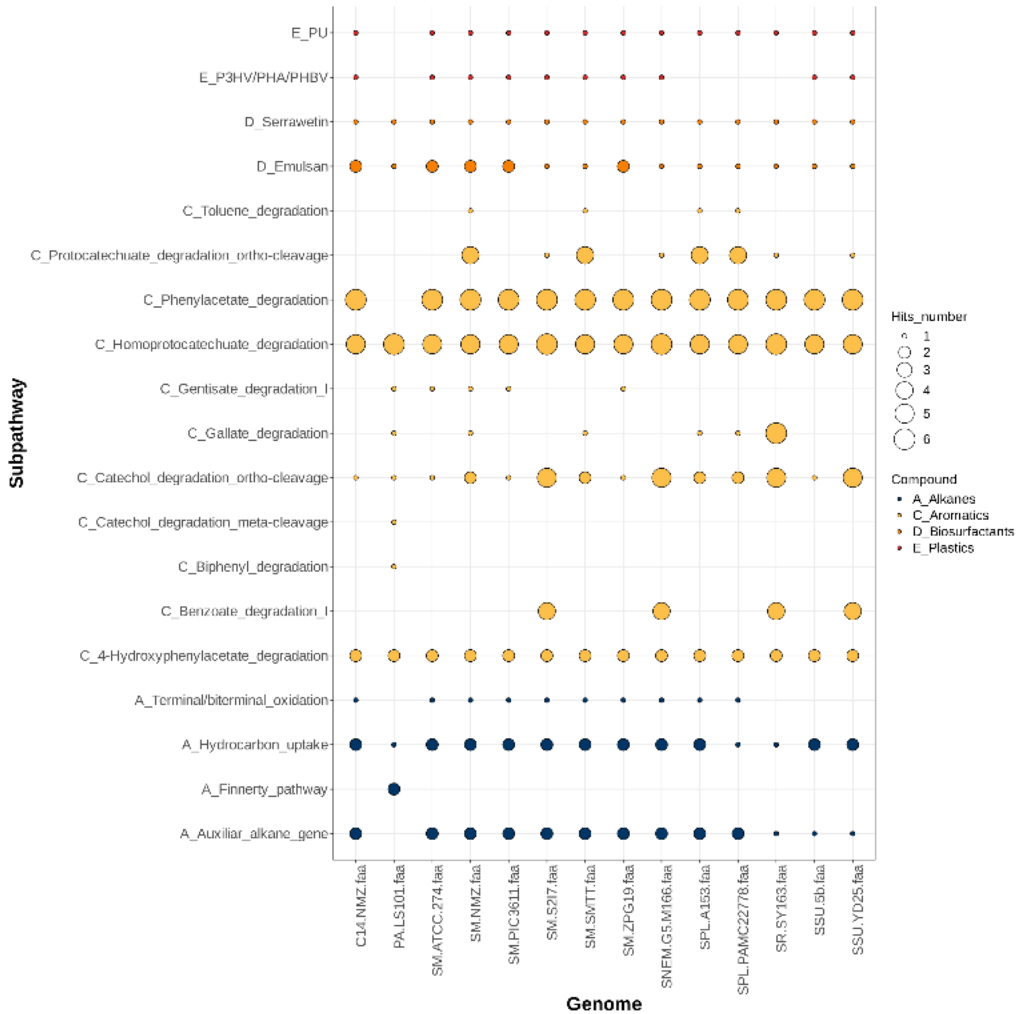
Reading the SwrW neighborhood



Arrows
Gene length and direction

Ribbons
Protein matches

Reading HADEG marker categories



Emulsan
Generic capsule markers

Toluene_degradation
Hydroxybenzoate-hydroxylase candidate

Bubble counts describe markers.
They do not measure pathway activity.

Chemical assignments and peak areas

Observation	What it can support
Exact mass and adduct	A candidate molecular composition
Fragmentation and reference match	Additional structural evidence
Authentic standard, when documented	A stronger identity comparison
Relative peak area	Detector signal under a stated denominator

The source reports chemical assignments. Genomics adds biosynthetic context.

Additional SM aromatic candidates

C SM-NMZ hydroxybenzoate-hydroxylase / pcaK neighborhood

SM-NMZ | 2,461,766–2,467,540 bp | NNLMDO_011760–011775



D SM-NMZ protocatechuate-associated region

SM-NMZ | 3,121,817–3,130,237 bp | NNLMDO_014965–015005

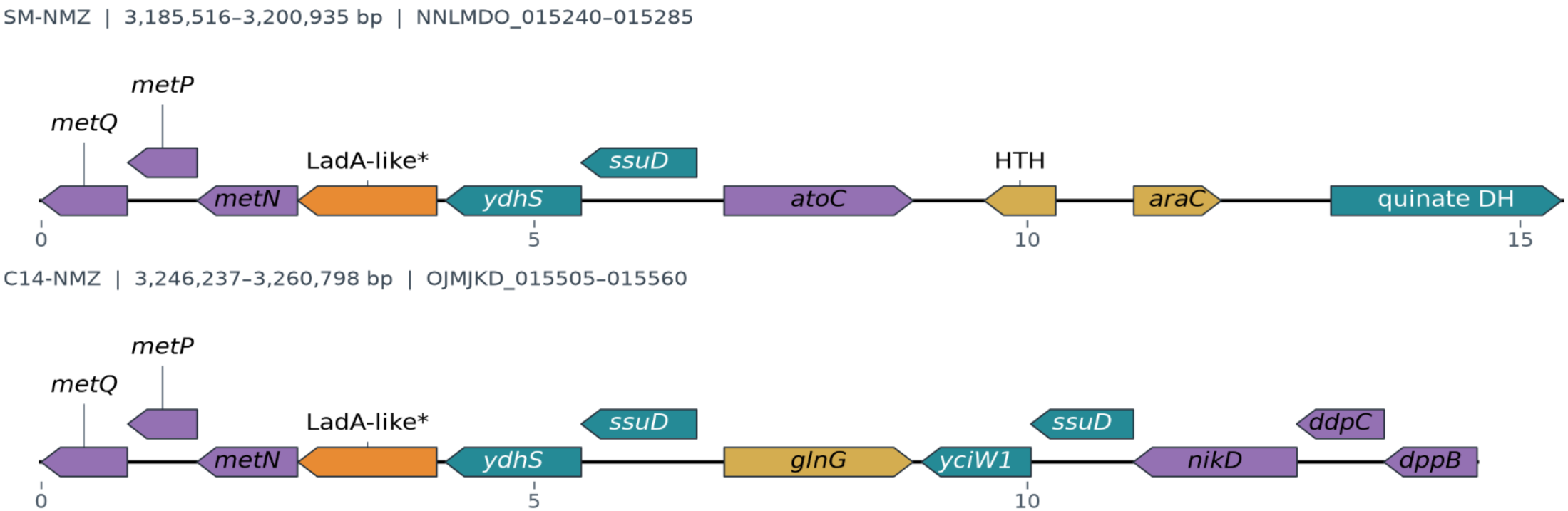


Scale in kb within each panel. Panels use different physical scales.

Additional candidates broaden the substrate hypotheses.

Some downstream steps remain unresolved.

LadA-like genomic neighborhoods



Gene arrows. Scale in kb from the first displayed CDS.

Source gene qualifier: ssuD. Candidate substrate remains unresolved.